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MUSings on a Crohn's Disease Diagnosis

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Keywords

CMUSE; cryptogenic multifocal ulcerative stenosing enteritis; small bowel ulcers; chronic ulcerative conditions of the small bowel; enteropathy; IBD mimic

Background

A 66-year-old male was evaluated at the University of North Carolina Multidisciplinary IBD clinic for suspected Crohn's disease (CD). His initial encounter in this clinic was preceded by over five years of mild generalized abdominal discomfort that had been attributed to irritable bowel syndrome. Over the year preceding his initial presentation to clinic, he developed weight loss, nausea, and bloating that persisted despite treatment for proven *Helicobacter pylori* infection. He had never undergone abdominal surgery, had no family history IBD, and denied smoking, alcohol and NSAID use.

Clinical Course

At the time of his presentation, the only significant abnormalities in his labs included findings consistent with iron deficiency anemia (Hgb 10.3 g/dL [normal 13–17.7 g/dL], ferritin 10 ng/mL [normal 10.5–307.3 ng/mL]), and hypoalbuminemia (2.7g/dL [normal 3.4–5.0g/dL]). Notably, he had normal systemic inflammatory markers (CRP 6.9mg/L [normal ≤10mg/L], ESR 4mm/h [normal 0–20mm/h]), and vitamin B12 (979pg/mL [normal 211–911pg/mL]). After his initial visit to the IBD clinic, he underwent an esophagogastroduodenoscopy and colonoscopy which were notable for duodenal ulcerations and a duodenal stenosis measuring <2cm that was traversable with an upper endoscope. Endoscopic biopsies showed mild to moderate enteritis without convincing evidence of chronicity, without granulomas, and were negative for dysplasia. A subsequent small bowel follow-through examination demonstrated multiple sites of narrowing throughout the entire small bowel. The patient was asked again about NSAID use, which he denied. He had no risk factors for tuberculosis. Based on the overall picture, he was diagnosed with mild small bowel CD and started on mesalamine.

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In subsequent years, he continued to have recurrent abdominal pain, bloating, and nausea. His therapy was escalated to adalimumab, with no improvement. He was then started on vedolizumab, which improved his abdominal discomfort. Unfortunately, due to the development a severe drug-induced rash, he was ultimately switched to ustekinumab in combination with azathioprine. Throughout this time, he was also intermittently successfully treated with budesonide for symptom exacerbations. Upper endoscopy and colonoscopy continued to reveal numerous (at least 4) short-segment (<2cm) duodenal and ileal strictures with shallow ulcerations and minimal inflammation surrounding areas of stricturing that were endoscopically dilated with variable clinical response (Figure 1). Biopsies continued to show mild-to-moderate enteritis. Over 6–7 years, the patient had yearly CT and MR enterographies which continued to show numerous (>10) short segment strictures throughout his small bowel with varying degrees of proximal dilation, measuring up to 4.5cm (Figure 2). Laboratory values continued to demonstrate chronic iron deficiency anemia and a normal C-reactive protein. He developed multiple small bowel obstructions requiring an exploratory laparotomy, which revealed multiple stenoses of his small bowel not amenable to strictureplasty due to the excessive number of stenoses. During a second laparotomy for high-grade bowel obstruction, he was also noted to have a perforation proximal to a distal ileal stricture that required resection with diverting ileostomy, with subsequent takedown. Histology from the ileal resection specimen revealed mild ileitis with surface adhesions and histologic evidence of perforation without malignancy or dysplasia. He was treated with short-term total parenteral nutrition (TPN) during this time.

Due to the numerous short-segment strictures, minimal to no systemic inflammation, absence of extra intestinal manifestations, and lack of response to conventional CD medications, his case was felt to be more consistent with cryptogenic multifocal ulcerative stenosing enteritis (CMUSE) rather than small bowel CD. He has been maintained on ustekinumab, azathioprine and budesonide for the past three years with stable control of his disease, including no significant bowel obstructions, no need for parenteral nutrition, and no need for repeat surgical intervention.

DISCUSSION

Chronic ulcerative conditions of the small intestine are a group of rare conditions of increasing importance, characterized by recurrent, superficial ulcerations and stenoses of the small bowel with negligible systemic inflammation. This term encompasses several diseases, including CMUSE and chronic nonspecific multiple ulcers of the small intestine (CNSU), later renamed chronic enteropathy associated with the SLC02A1 gene (CEAS).

There are currently no epidemiological data on incidence and prevalence of these conditions. Although they are unusual, improved sensitivity of radiological techniques and diagnostic capabilities are believed to increase their recognition in clinical practice. Their rarity is also due in part to misdiagnosis due to their similarity to more common conditions such as small bowel CD and NSAID enteropathy. Other conditions on the differential diagnosis include tuberculous enteritis, Behçet's disease, radiation enteritis, and adenocarcinomas or lymphomas. Moreover, as seen in our case, a key feature of CMUSE and CEAS is the finding of multiple short strictures in the small intestine associated with minimal,

non-specific inflammatory changes, including lack of granulomas and absence of transmural inflammation.

CMUSE was first described in a French case report in the 1960s [1]; subsequently, a large number of studies have been done in France to characterize the condition. Though the majority of studies recognizing and characterizing CEAS have been conducted in Japan [2], cases are currently recognized in numerous European and Asian countries as well as in the United States. The pathophysiology is unknown; some have speculated an underlying autoimmune and/or atypical vasculitis mechanism. Recent research suggest that CMUSE and CEAS are prostaglandin-mediated conditions, associated with loss-of-function mutations of genes for proteins involved in prostaglandin transport across the vascular endothelium [4]. More specifically, the solute cotransporters (SLC)02A1 and phospholipase A2 Group IV (PLA2G4) genes are implicated in CEAS and CMUSE, respectively, the former pair being more strongly associated. These genes are thought to be inherited in an autosomal recessive pattern [3–6]. Prostaglandins maintain microvascular blood flow and are important for the mucosal integrity of the gut.

Clinical features of CMUSE and CEAS include obstructive symptoms (i.e. abdominal pain, bloating, nausea, vomiting) and chronic anemia, with the potential of significant bleeding from ulcerations. Abscesses and fistulas seen in penetrating CD are typically absent. Nutritional complications and weight loss are commonly seen, and depending on the course of illness, individuals may require long-term TPN. The relapsing nature of these conditions often requires surgical intervention, though recurrence can occur even after surgery. Extra-intestinal manifestations are generally absent in CMUSE, whereas CEAS may have unique extra-intestinal symptoms related to the SLC02A1 mutation, such as primary hypertrophic osteoarthropathy, pachydermoperiostosis, and isolated nail clubbing [4].

Diagnosis is typically made from clinical history as well as imaging and endoscopy. Imaging includes CT or MR enterography. Though there are no formal diagnostic parameters, the presence of multiple (>5), short segment (<2cm), circumferential strictures in the absence of pathognomonic features of CD, NSAID use, and risk factors for alternative causes such as TB or lymphoma can be suggestive of CMUSE/CEAS [7]. Capsule endoscopy is discouraged when strictures are identified due to the high rate of capsule retention ($\leq 60\%$) [8]. Balloon enteroscopy facilitates both visualization and intervention. Endoscopy typically shows one or more strictures with shallow ulcers restricted to areas of stenosis in the small bowel with intervening normal mucosa. Although the jejunum and ileum are more commonly involved, disease can be present throughout the small bowel. Notably, involvement of the stomach or colon is not seen. Ulceration can vary in morphology – sometimes, these areas will have vascular features such as angioectasias. Biopsies show non-specific inflammatory infiltrates (i.e. neutrophils, lymphocytes) restricted to mucosa and submucosa. Unlike CD, there is no transmural involvement and no granuloma formation [4].

Effective medical treatment options for these conditions are lacking. CD-directed therapies are often used in practice though supporting evidence is limited. Systemic and local corticosteroids are often initially tried, though with mixed results. Although there have been case reports of immune modulators and infliximab inducing clinical remission, the

lack of robust data and long-term follow-up preclude their routine use [9]. In cases associated with angioectasias, estradiol and norethindrone reduce recurrent gastrointestinal hemorrhage. Prostaglandin analogues such as misoprostol have shown some efficacy in case reports though further studies are needed to determine their use on a larger scale. Consumption of a low-residue diet and nutritional support including consideration of TPN are important aspects of therapy as well. Surgery is typically restricted to those with small bowel obstructions or those not responding to medical therapy; nevertheless, as mentioned above, these conditions often relapse after surgery. Although there is no consensus on how often to repeat endoscopy to manage strictures or monitor for disease, it is advised to repeat enteroscopy during or after medical treatment to evaluate for mucosal remission and estimate the risk of relapse [4,5].

CMUSE and other chronic ulcerative conditions of the small bowel are a distinct spectrum of conditions that all gastroenterologists should be aware of and consider in patients with ulcerative and stricturing disease isolated to the small bowel. This case highlights features that can aid in the identification of these conditions and in the differentiation from CD, and also demonstrates the challenges of managing severe, refractory disease, emphasizing the importance of multidisciplinary care, individualized treatment strategies, and close monitoring for complications.

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Conflicts of Interest:

Millie D. Long has served as a consultant for AbbVie, Pfizer, Takeda, Lilly, Bristol-Meyers Squibb, Target RWE, and Prometheus and has received research support from Pfizer, Takeda and Janssen.

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REFERENCES

1. Debray C et al. Ent'rite st'enosante ulc'ereuse plurifocale cryptog'en'etique [cryptogenetic plurifocal ulcerative stenosing enteritis]. *Arch mal appar dig mal nutr.* 1964;53:193–206. [PubMed: 14140712]
2. Umeno J, Esaki M, Hirano A, et al. Clinical features of chronic enteropathy associated with SLCO2A1 gene: a new entity clinically distinct from Crohn's disease. *J Gastroenterol.* 2018;53(8):907–915. doi:10.1007/s00535-017-1426-y [PubMed: 29313109]
3. Perlemuter G, Guillevin L, Legman P, Weiss L, Couturier D, Chaussade S. Cryptogenetic multifocal ulcerous stenosing enteritis: an atypical type of vasculitis or a disease mimicking vasculitis. *Gut.* 2001;48(3):333–338. doi:10.1136/gut.48.3.333 [PubMed: 11171822]
4. Singh A. Cryptogenic Multifocal Ulcerating Stenosing Enteropathy(CMUSE) and/or Chronic Non-specific Multiple Ulcers of the Small Intestine(CNSU) and Non-granulomatous Ulcerating Jejunoileitis (NGUIJ). *Curr Gastroenterol Rep.* 2019;21(10):53. Published 2019 Sep 10. doi:10.1007/s11894-019-0721-6 [PubMed: 31501950]

5. Moreels TG, Singh A. Updates on the diagnosis and management of cryptogenic multifocal ulcerative stenosing enteropathy (CMUSE) and non-steroidal enteropathy. *Best Pract Res Clin Gastroenterol*. 2023;64–65:101847. doi:10.1016/j.bpg.2023.101847
6. McAlindon Mark E.. Cryptogenic multifocal ulcerating stenosing enteritis and other under-recognised small bowel inflammatory enteropathies. *Current Opinion in Gastroenterology* 38(3):p 279–284, May 2022. | DOI: 10.1097/MOG.0000000000000833 [PubMed: 35275899]
7. Ramos GP, Bartlett DJ, Bledsoe AC, et al. Cryptogenic multifocal ulcerous stenosing enteritis (CMUSE): a 20-year single-center clinical and radiologic experience. *Abdom Radiol (NY)*. 2021;46(8):3798–3809. doi:10.1007/s00261-021-03005-y [PubMed: 33728531]
8. Tao EW, Zou TH, Wang YF, Tang JT, Chen YX, Gao QY. Case report of cryptogenic multifocal ulcerous stenosing enteritis (CMUSE): a rare disease may contribute to endoscopy-capsule retention in the small intestine [published correction appears in *BMC Gastroenterol*. 2019 Apr 29;19(1):64. doi: 10.1186/s12876-019-0983-3]. *BMC Gastroenterol*. 2019;19(1):49. Published 2019 Apr 3. doi:10.1186/s12876-019-0962-8 [PubMed: 30943900]
9. De Schepper H, Macken E, Van Marck V, Spinhoven M, Pelckmans P, Moreels T. Infliximab induces remission in cryptogenic multifocal ulcerous stenosing enteritis: first case. *World J Gastroenterol*. 2013 Mar 14;19(10):1661–4. doi: 10.3748/wjg.v19.i10.1661. [PubMed: 23539523]

KEY POINTS

- Chronic ulcerative and stricturing conditions of the small bowel (i.e. CMUSE, CEAS) are an increasingly recognized unusual but distinct spectrum of conditions that can mimic other inflammatory, infectious, or drug-induced intestinal ulcerating strictures.
- Consider CMUS or CEAS when there are numerous, short-segment, circumferential strictures associated with superficial ulcerations isolated to the small bowel and with minimal/no systemic inflammation.
- Key differentiating factors from Crohn's disease are a lack of systemic inflammation, lack of gastric and colonic involvement, lack of transmural inflammation and granulomas, and generally poor response to CD-directed therapies.

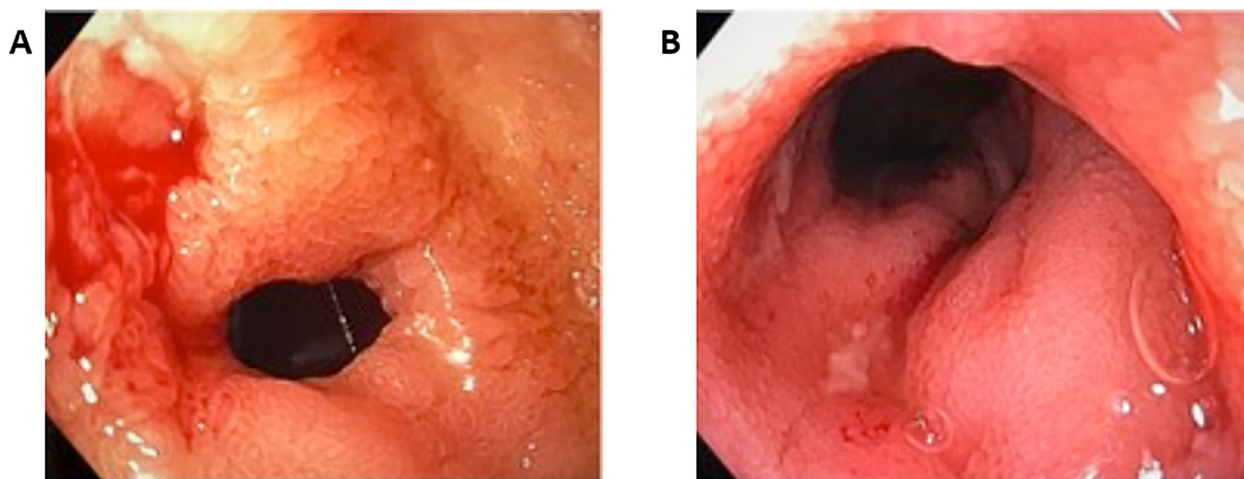


Figure 1. Endoscopic images illustrating strictures with associated ulceration in the duodenum (A) and ileum (B).

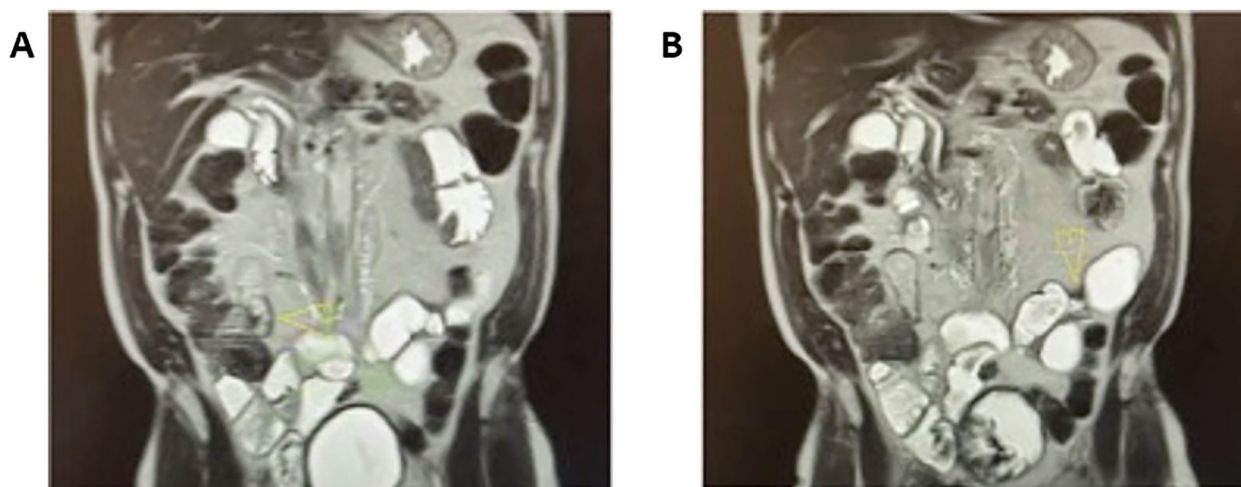


Figure 2.
MR Enterography illustrating strictures in the small bowel.

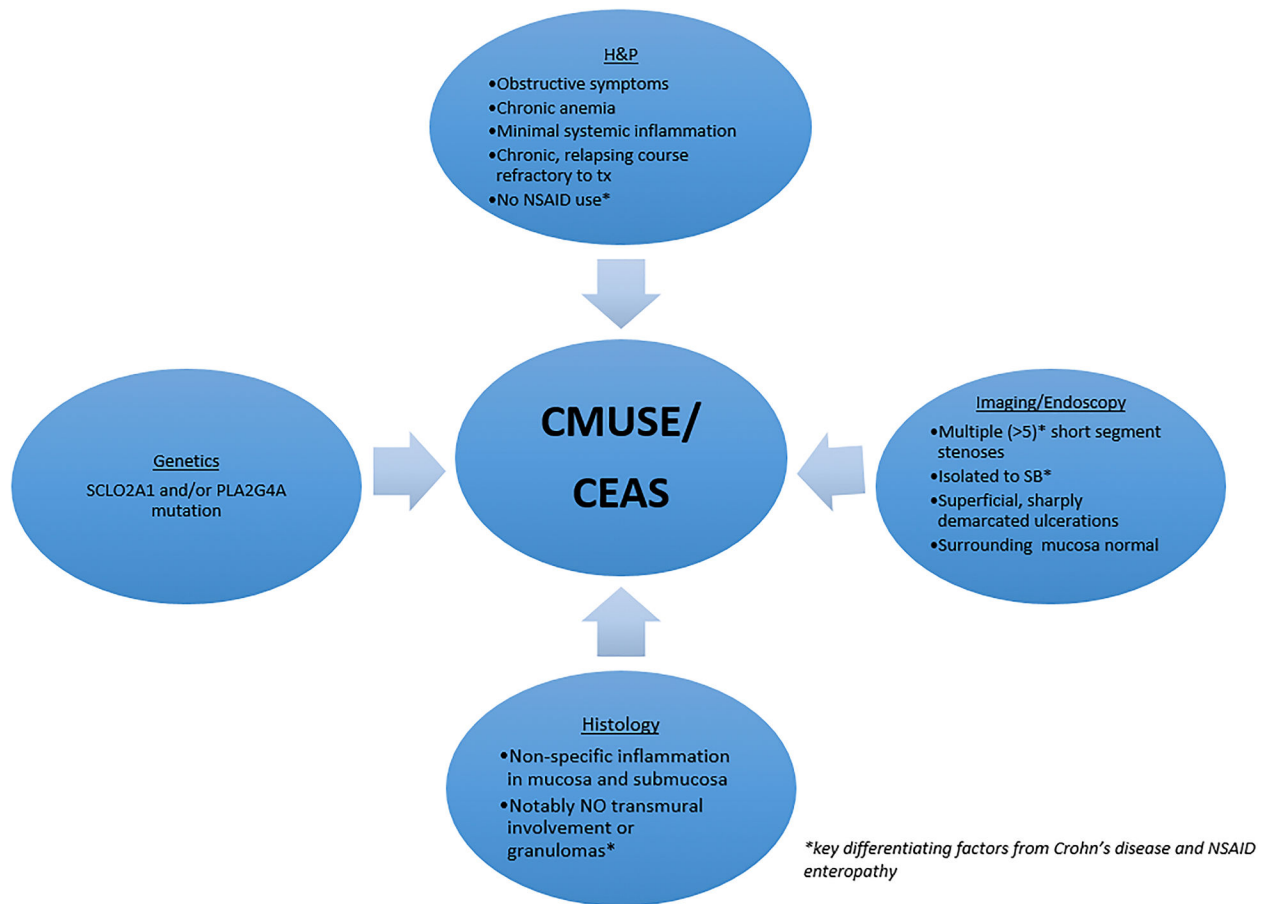


Figure 3. Considerations in the evaluation of a patient with suspected chronic ulcerative conditions of the small bowel.